

Year Two Lecture Notes

BSc Computer Science - Durham University

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October 2026

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Part I

Artificial Intelligence

1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

This module is assessed through three pieces of summative coursework, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	10/10/2026	16/12/2026	22/01/2026	Dr Yang Long	50%
AI Search	12/01/2027	19/02/2027	19/03/2026	Prof. Iain Stewart	25%
Bias in AI	06/02/2027	26/03/2027	07/05/2027	Dr. Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	n/a
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, Faisal and Ong, 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, 2017).

2.1 Introduction to Machine Learning

2.2 Machine Learning Workflow and Coding Environment

2.3 Data Understanding

2.4 Data Preparation Workflow

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.3 General Heuristic Search: Greedy Best-First Search and A* Search

3.4 A* Search and Deriving Iterative Deepening A* Search

3.5 Heuristic Functions and Their Development

3.6 Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms

3.7 Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm

3.8 Applying Different Ant Systems to the TSP

3.9 General Versions of Particle Swarm Optimisation and Differential Evolution

3.10 Applying Particle Swarm Optimization to the TSP

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

4.1 Introduction to AI Ethics and Related Concepts

4.2 Societal Impacts of AI

4.3 Defining Bias and Fairness in ML

4.4 How to Build Better Systems

4.5 Value Sensitive Design

4.6 Evaluations and Audits

4.7 Bias Mitigation Strategies

4.8 Transparency and Accountability for Trustworthy Systems

4.9 Explainable AI

4.10 Future Trends and Ethical Frontiers

Part II

Data Science

5 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

5.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 3. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Data Cleaning & Analytics	Week 11	n/a	13/02/2027	Dr Igor Razgon	25%
Image Processing	16/01/2027	26/02/2027	26/03/2027	Dr George Kouliris	25%
Probability and Graphics	May Exam	n/a	n/a	Razgon & Kouliris	50%

Table 3: Assessments for Data Science.

5.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 4.

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	n/a

Table 4: Michaelmas Term Timetable for Data Science.

6 Probability and Statistics

This section is taught by Dr Igor Razgon. There is no recommended reading for this topic.

- 6.1 Basics of Counting and Probability
 - 6.2 Dependence, Conditional Probability, and Bayes' Theorem
 - 6.3 Discrete Random Values and Probability Distributions
 - 6.4 Continuous Probability Distributions
 - 6.5 The Normal Distribution
 - 6.6 Multivariate Distributions
 - 6.7 Statistics and the Central Limit Theorem
 - 6.8 Point Estimators, Confidence Intervals, and Hypothesis Testing
 - 6.9 Hypothesis Testing
 - 6.10 From Statistics to Data
- ## 7 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon. [Check for recommended reading.](#)

8 Computer Graphics

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

9 Image Processing

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

Part III

Networks and Systems*

10 Module Introduction

Networks and Systems has the module code COMP2211 and is led by Dr Wanqing Tu.

10.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 5. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Cyber Security	10/10/2026	06/11/2026	04/12/2026	Dr Charles Morisset	25%
Networks	31/10/2026	08/01/2027	05/02/2027	Dr Wanqing Tu	25%
Dist. Sys. & Databases	May Exam	n/a	n/a	To be confirmed.	50%

Table 5: Assessments for Networks and Systems.

10.2 Timetable

This module is taught through one two-hour lecture and one two-hour computer class each week as seen below in Table 6. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Thursday	09:00 - 11:00	Lecture	CLC202	Dr Carlton Shepherd
Thursday	16:00 - 18:00	Computer Class	CC-0007	Dr Carlton Shepherd

Table 6: Michaelmas Term Timetable for Networks and Systems.

11 Cyber Security

This section is taught by... **Check for recommended reading.**

12 Networks

This section is taught by... **Check for recommended reading.**

13 Distributed Systems

This section is taught by... **Check for recommended reading.**

14 Databases

This section is taught by... **Check for recommended reading.**

Part IV

Programming Paradigms

15 Module Introduction

Programming Paradigms has the module code COMP2221 and is led by Dr Stuart James.

15.1 Assessment

This module is assessed through one piece of summative coursework and two final exams, these can be seen in Table 7. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Systems Programming	31/10/2026	11/12/2026	08/01/2027	Dr Stuart James	50%
Object Oriented & Functional	May Exam	n/a	n/a	Ivkovic & Moses	50%

Table 7: Assessments for Programming Paradigms.

15.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 8.

Day	Time	Class Type	Location	Lecturer
Monday	11:00 - 12:00	Lecture	CLC202	Dr Stuart James
Monday	16:00 - 18:00	Computer Class	CB-LG001	n/a
Friday	14:00 - 15:00	Lecture	CLC202	Dr Stuart James

Table 8: Michaelmas Term Timetable for Programming Paradigms.

16 Systems Programming

This section is taught by Dr Stuart James. **Check for recommended reading.**

17 Functional Programming

This section is taught by Dr Billy Moses Jr.. **Check for recommended reading.**

18 Object Oriented Programming

This section is taught by Dr Mladen Ivkovic. There is no recommended reading for this topic.

- 18.1 Java Fundamentals**
- 18.2 OOP with Classes and Objects**
- 18.3 Encapsulation**
- 18.4 Inheritance**
- 18.5 Abstraction with Abstract Classes**
- 18.6 Abstraction with Interfaces**
- 18.7 Polymorphism**
- 18.8 Java Collections**
- 18.9 Best Practises**

Part V

Software Engineering

19 Module Introduction

Software Engineering has the module code COMP2281 and is led by Dr Craig Stewart.

19.1 Assessment

This module is assessed through a group project, the deadlines for this can be seen in Table 9. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Peer Evaluation 1	17/10/2026	20/11/2026	22/01/2027	Dr Craig Stewart	1.67%
Requirement Documentation	17/10/2026	20/11/2026	22/01/2027	Dr Craig Stewart	15%
Peer Evaluation 2	17/10/2026	12/02/2027	07/05/2027	Dr Craig Stewart	1.67%
Technical Report	17/10/2026	12/03/2027	07/05/2027	Dr Craig Stewart	30%
Product Presentation	17/10/2026	20/03/2027	07/05/2027	Dr Craig Stewart	15%
Peer Evaluation 3	17/10/2026	24/03/2027	07/05/2027	Dr Craig Stewart	1.67%
Reflective Experience Report	17/10/2026	30/04/2027	n/a	Dr Craig Stewart	20%
Project Handover	17/10/2026	30/04/2027	n/a	Dr Craig Stewart	10%
Git Commit	17/10/2026	30/04/2027	n/a	Dr Craig Stewart	5%

Table 9: Assessments for Software Engineering.

19.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 10. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Tuesday	09:00 - 10:00	Lecture	TLC033	Dr Nelly Bencomo
Wednesday	11:00 - 12:00	Lecture	TLC033	Dr Nelly Bencomo
Thursday	11:00 - 13:00	Practical	CC-0007	Dr Gagangeet Aujla

Table 10: Michaelmas Term Timetable for Software Engineering.

20 Planning and Design

20.1 Requirement Engineering

20.2 Modelling and Analysis

20.3 Software Design

21 Development and Testing

21.1 Software Development Methodologies

21.2 Software Testing

22 Development Considerations

22.1 Software Quality Management

22.2 Human Computer Interaction

22.3 Legal, Social, Ethical, and Professional Issues

23 Group Project

23.1 Introduction

23.2 Project Management

Part VI

Theory of Computation

24 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

24.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 11.

Title	Opening	Due	Feedback	Professor	Weight
Benchtest 1	Week 11	n/a	09/02/2027	Dr Dantchev and Dr Mertzios	17%
Benchtest 2	Week 20	n/a	07/05/2027	Dr Dantchev and Dr Mertzios	17%
Final Exam	May Exam	n/a	n/a	Dr Barnaby Martin	66%

Table 11: Assessments for Theory of Computation.

24.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 12.

Day	Time	Class Type	Location	Lecturer
Tuesday	10:00 - 11:00	Lecture	CLC202	Dr Barnaby Martin
Wednesday	09:00 - 11:00	Problem Class	TLC116	n/a
Thursday	14:00 - 15:00	Lecture	CLC202	Dr Sukanya Pandey

Table 12: Michaelmas Term Timetable for Theory of Computation.

25 Models of Computation

This section is taught by Dr Barnaby Martin.

25.1 Finite-State Automata and Regular Languages

25.2 Büchi Automata and Omega-Regular Languages

25.3 Linear-Time Logic and Translating Into Büchi Automata

26 Algorithms and Complexity

26.1 Asymptotic Complexity and Notation

26.2 Greedy Algorithms

26.3 Divide and Conquer Algorithms

26.4 Dynamic Programming Algorithms

26.5 Machinings in Graphs

Part VII

Appendix

27 Artificial Intelligence

27.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

27.2 References

27.3 List of Figures

28 Reading List

28.1 Machine Learning

Deisenroth, M.P., Faisal, A.A. and Ong, C.S. (2020). Mathematics for Machine Learning. [online] Cambridge University Press. Available at: <https://mml-book.com/> [Accessed 3 Oct. 2026].

Hackeling, G. (2017). Mastering Machine Learning with scikit-learn. 2nd edn. Packt Publishing.

28.2 Ethics and Bias in AI

Various, will be added over time.

29 Data Science

29.1 Module Professors

Module Lead: Professor Thoas Erlebach

- Focus: Algorithms and networks.

- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00.

Lecturer: Doctor George Alex Koulieris

- Focus: Image processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

29.2 References

29.3 List of Figures

30 Networks and Systems

30.1 Module Professors

Module Lead: Doctor Wanqing Tu

- Focus: Computer networks and wireless communications.
- Email: wanqing.tu@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Charles Morisset

- Focus: Explainable security, access control, and the security and privacy of smart infrastructure.
- Email: charles.morisset@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Carlton Shepherd

- Focus: Cyber security.
- Email: carlton.g.shepherd@durham.ac.uk
- Office Hours: **To be confirmed.**

30.2 References

30.3 List of Figures

31 Programming Paradigms

31.1 Module Professors

Module Lead: Doctor Stuart James

- Focus: Artificial intelligence, computer vision, human computer interaction, and digital humanities.
- Email: stuart.a.james@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Billy Moses Jr.

- Focus: Algorithms and biologically inspired computing.
- Email: william.k.moses-jr@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Mladen Ivkovic

- Focus: Performance prediction and optimisation.
- Email: mladen.ivkovic@durham.ac.uk
- Office Hours: Email to make an appointment.

31.2 References

31.3 List of Figures

32 Software Engineering

32.1 Module Professors

Module Lead: Doctor Craig Stewart

- Focus: Adaptive systems, HCI, and user modelling.
- Email: craig.d.stewart@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Nelly Bencomo

- Focus: Software engineering for autonomous, self-adaptive, AI and ML systems.
- Email: nelly.bencomo@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Gagangeet Aujla

- Focus: Data-driven and data-intensive cyber-physical systems and digital twins.
- Email: gagangeet.s.aujla@durham.ac.uk
- Office Hours: **To be confirmed.**

32.2 References

32.3 List of Figures

33 Theory of Computation

33.1 Module Professors

Module Lead: Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Sukanya Pandey

- Focus: Not specified.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Stefan Dantchev

- Focus: Combinatorics.
- Email: s.s.dantchev@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

33.2 References

33.3 List of Figures

33.4 Recommended Reading

Cormen, T.H., Leiserson, C.E, Rivest, R.L, and Stein, C. (2009) *Introduction to Algorithms - Third Edition*. MIT Press.